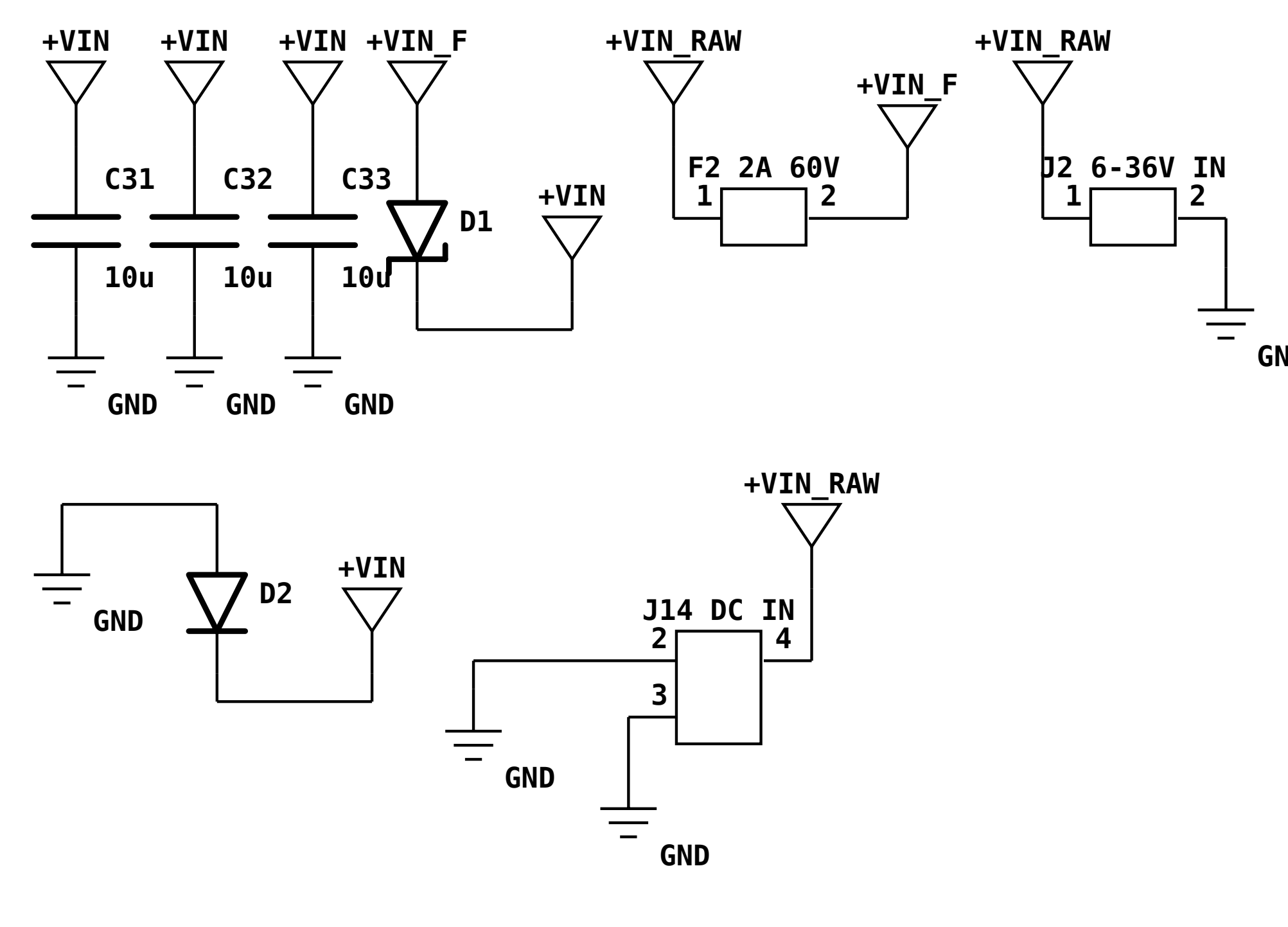
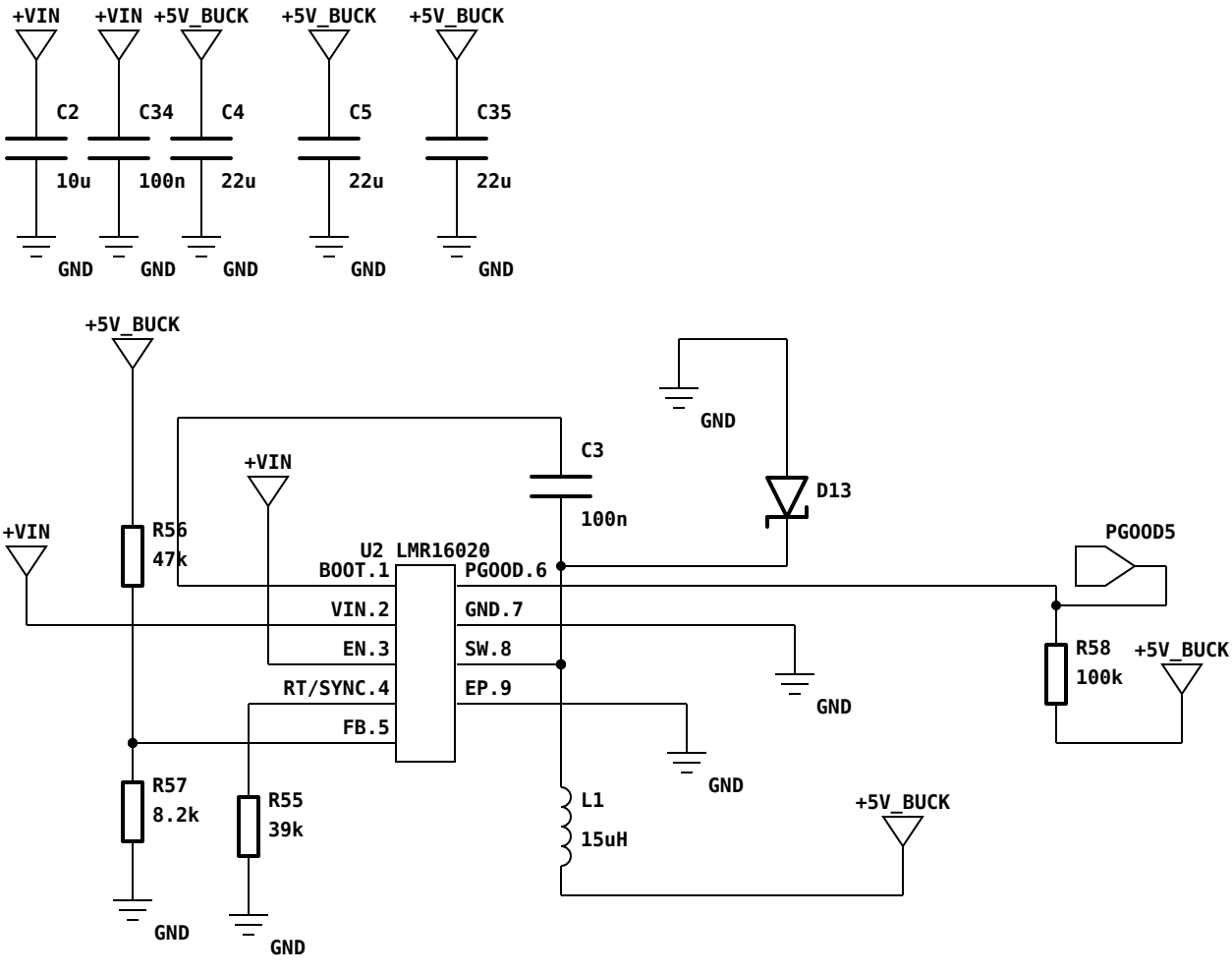
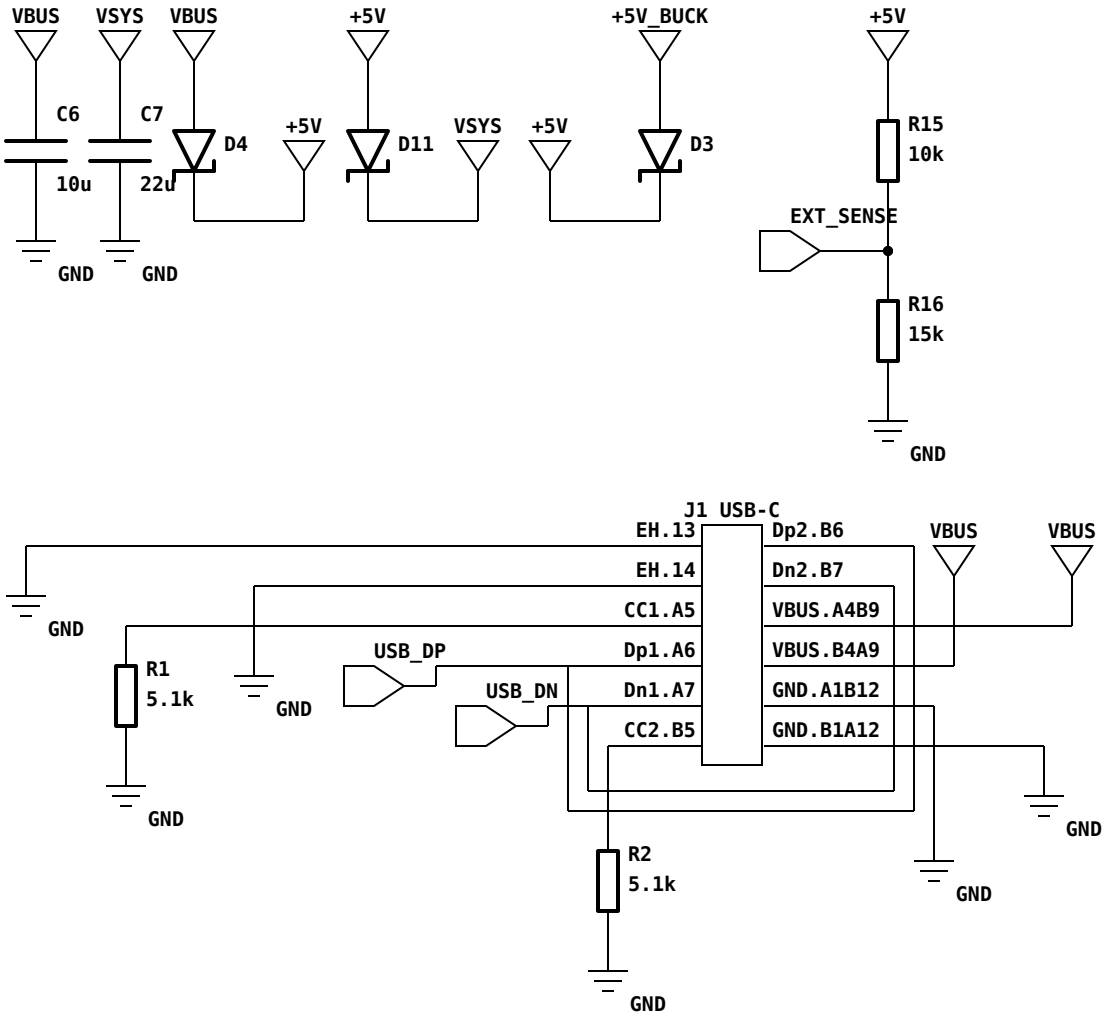
# Wiegand reader bridge — revision 11 schematic

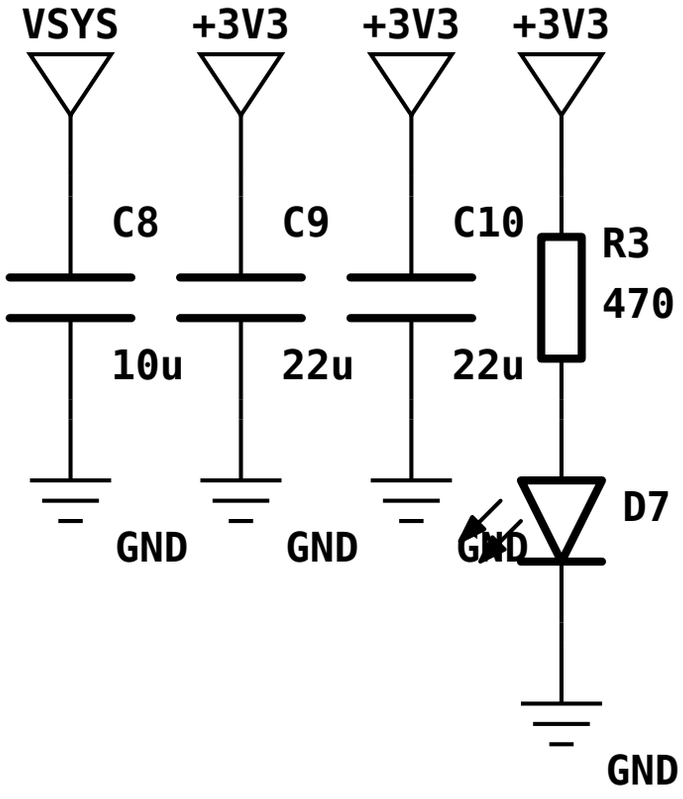
Generated from netlist.py (netlistsvg). Supply nets are flags; nets shared between blocks are named ports. I

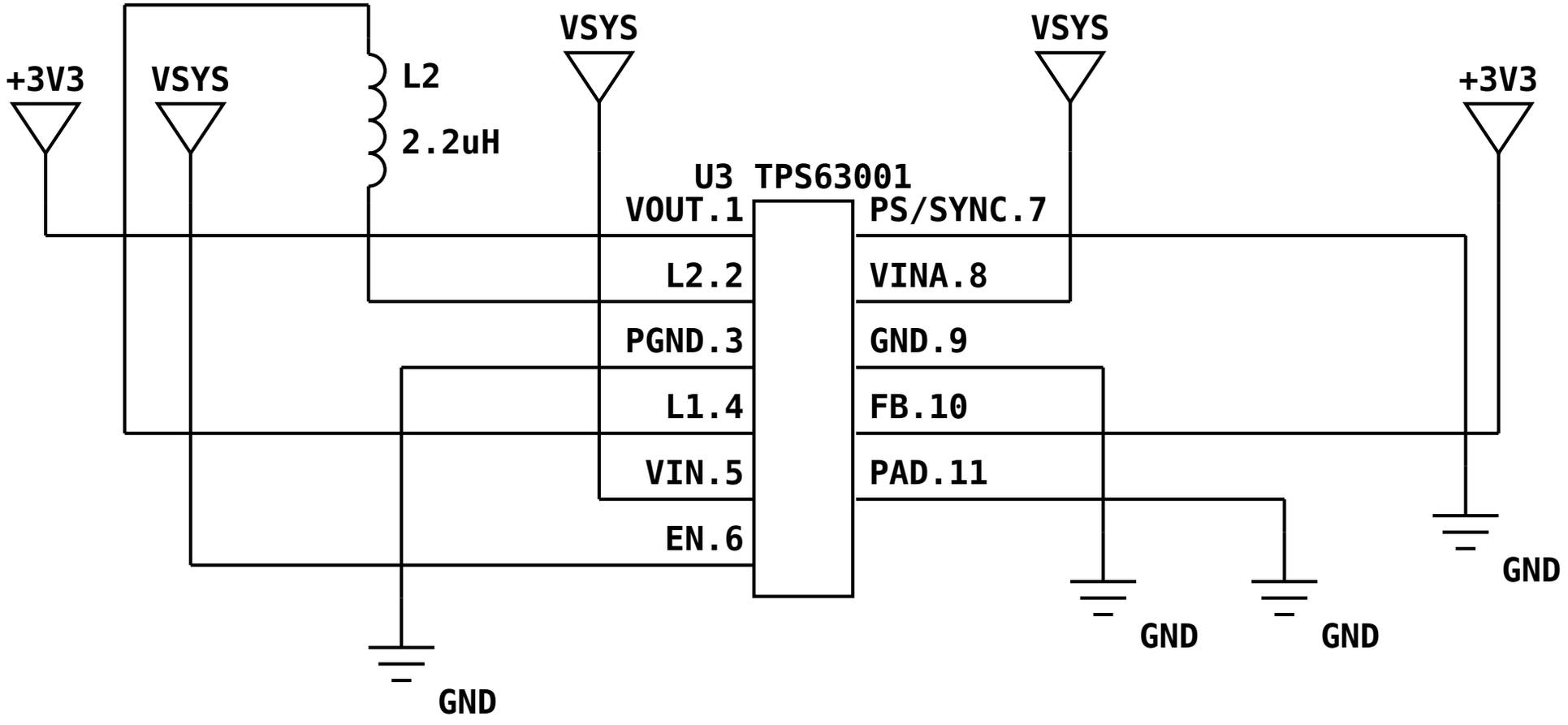
1. Block 1: 6-36 V input
2. Block 2: buck to 5 V (LMR16020, 60 V; datasheet SNVSAH8A sec. 8.2)
3. Block 3: USB-C, +5V, VSYS
4. Block 4: 3.3 V buck-boost
5. Block 5: MCU
6. Block 6: Wiegand inputs
7. Block 7: LED and BEEP
8. Block 8: reader supply
9. Block 9: battery
10. Block 10: screens
11. Block 11: inputs
12. Block 12 (rev 10): I/O expander, TCA9535 at 0x20
13. Block 13 (rev 10): relay -- driver fitted, relay and terminals NOT fitted
14. Block 14 (rev 10): RS-485 / OSDP on UART0
15. Block 15 (rev 10): surge/ESD on every field wire (SM712: +12 / -7 V standoff)
16. Block 16 (rev 12): reader monitor, INA226 on I2C (0x40)
17. Block 17 (rev 10): W5500 module header and Qwiic, not fitted

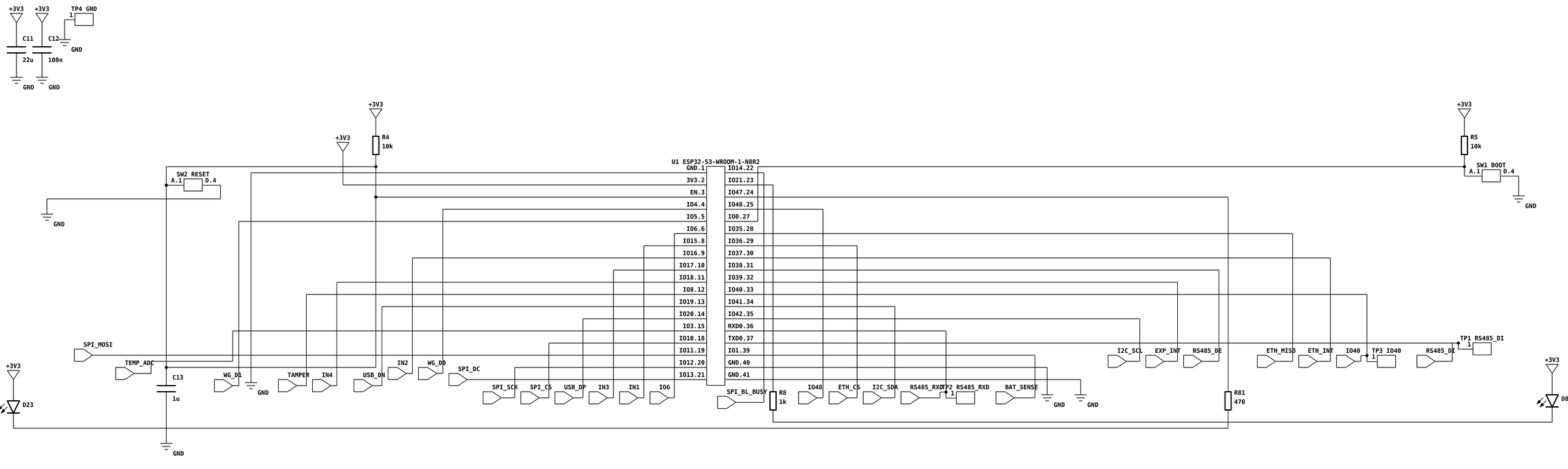

+VIN
+VIN
+VIN
+VIN_F
C31
C32
C33
10u
10u
10u
D1
+VIN
GND
GND
GND
+VIN_RAW
+VIN_F
F2 2A 60V
1
2
+VIN_RAW
J2 6-36V IN
1
2
GN
GND
D2
+VIN
+VIN_RAW
J14 DC IN
2
4
3
GND
GND

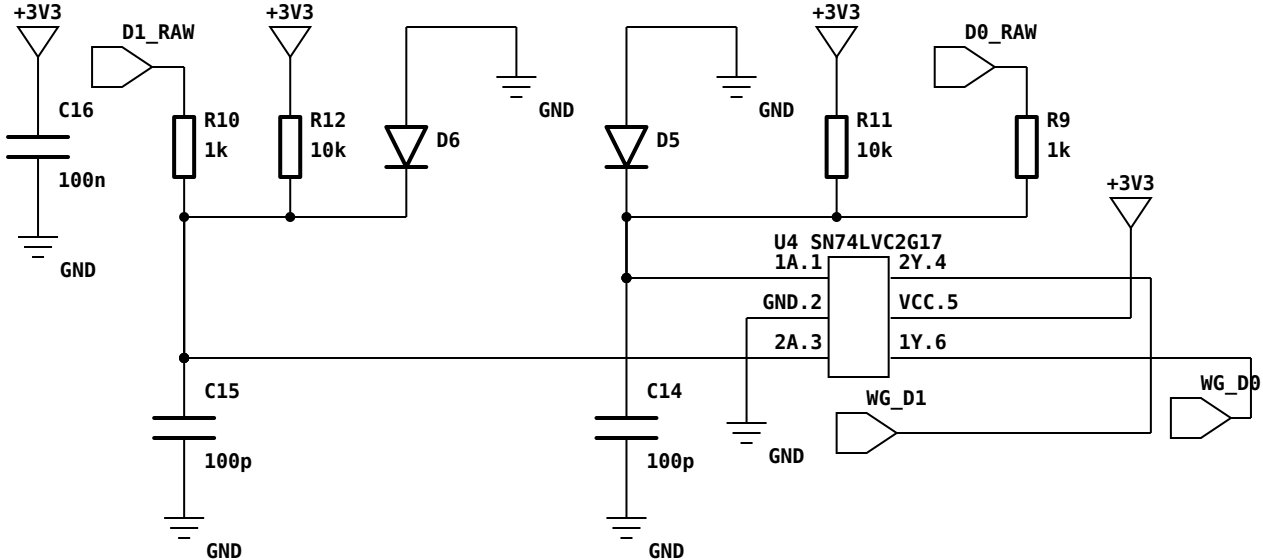

+VIN
+VIN
+5V_BUCK
+5V_BUCK
+5V_BUCK
C2
C34
C4
C5
C35
10u
100n
22u
22u
22u
GND
GND
GND
GND
GND
+5V_BUCK
+VIN
+VIN
R56
47k
C3
100n
GND
D13
U2 LMR16020
BOOT.1
VIN.2
EN.3
RT/SYNC.4
FB.5
PGOOD.6
GND.7
SW.8
EP.9
PGOOD5
R58
100k
+5V_BUCK
GND
GND
R57
8.2k
R55
39k
L1
15uH
+5V_BUCK
GND
GND

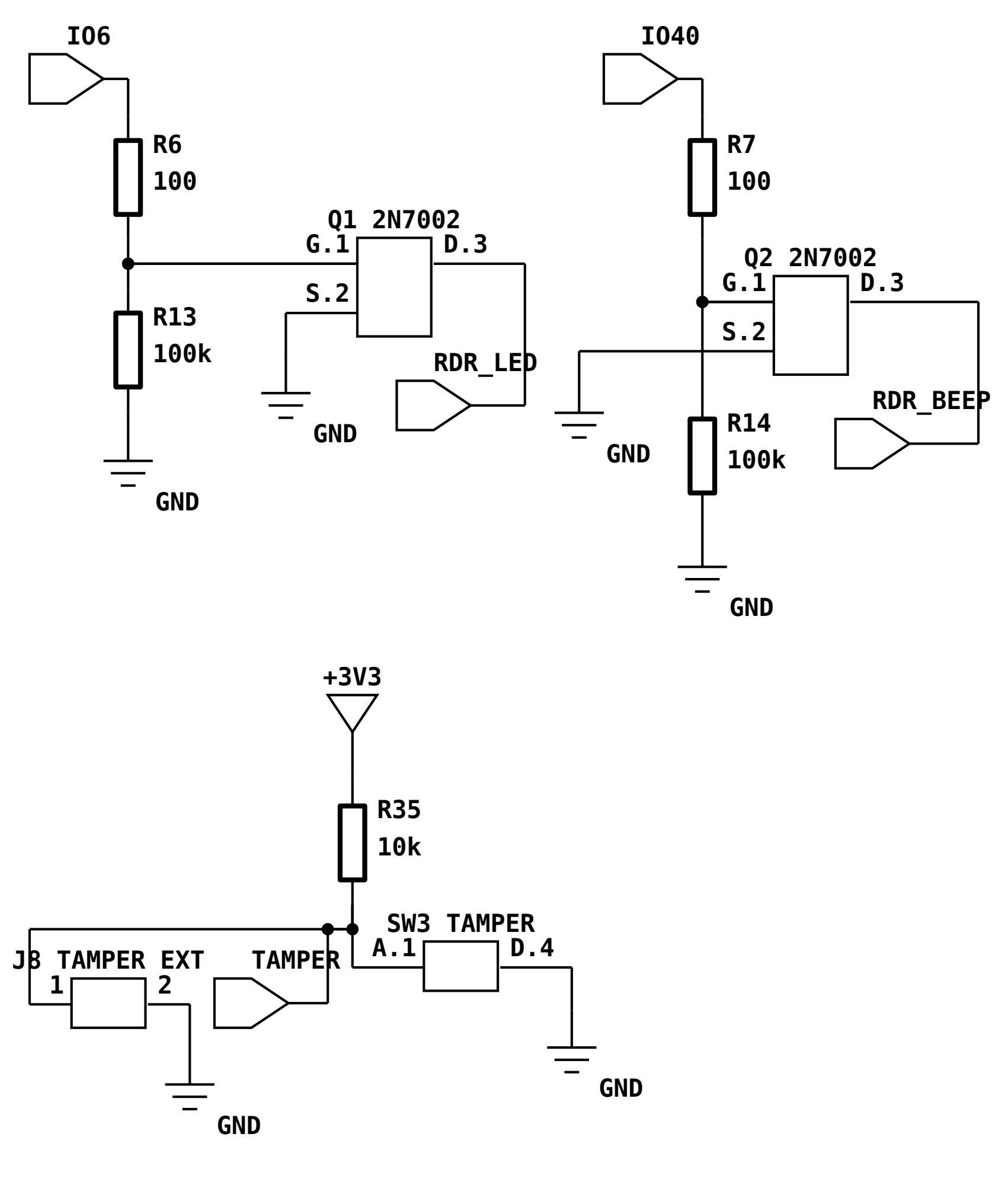

VBUS
VSYS
VBUS
+5V
+5V_BUCK
+5V
C6
10u
C7
22u
D4
+5V
D11
VSYS
+5V
D3
R15
10k
EXT_SENSE
R16
15k
GND
GND
GND
GND
J1 USB-C
EH.13
EH.14
CC1.A5
Dp1.A6
Dn1.A7
CC2.B5
Dp2.B6
Dn2.B7
VBUS.A4B9
VBUS.B4A9
GND.A1B12
GND.B1A12
VBUS
VBUS
GND
R1
5.1k
GND
USB_DP
USB_DN
GND
GND
GND
R2
5.1k
GND

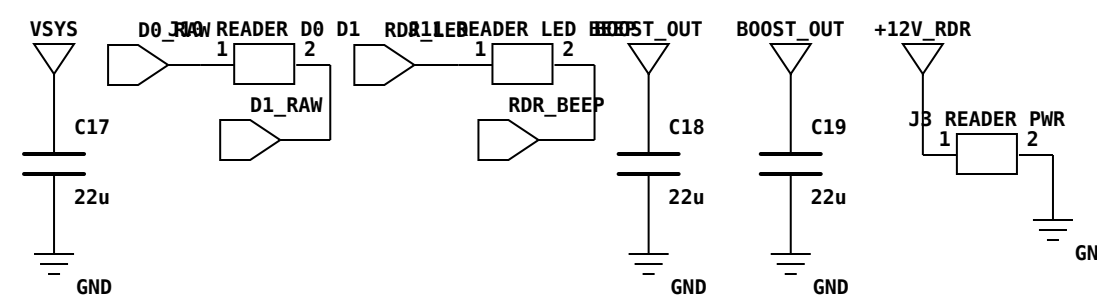

VSYS
+3V3
+3V3
+3V3
C8
C9
C10
R3
470
10u
22u
22u
GND
GND
GND
D7
GND

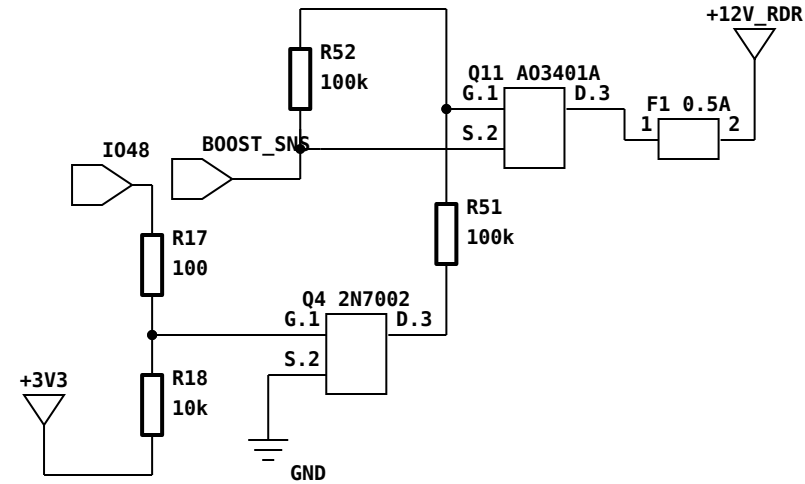

+3V3
VSYS
L2
2.2uH
VSYS
VSYS
+3V3
U3 TPS63001
VOUT.1
PS/SYNC.7
L2.2
VINA.8
PGND.3
GND.9
L1.4
FB.10
VIN.5
PAD.11
EN.6
GND
GND
GND
GND

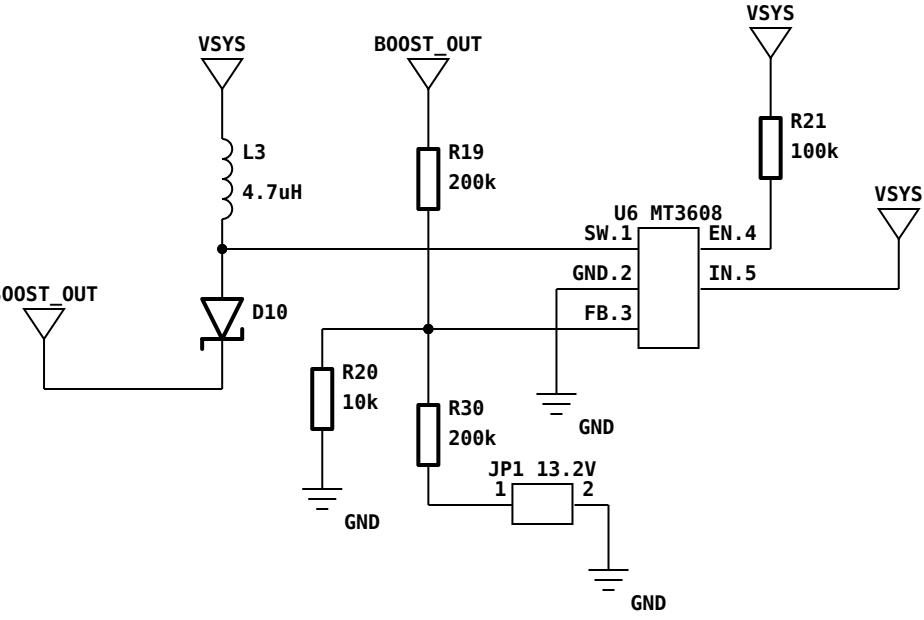

+3V3
+3V3
TP4 GND
1
C11
C12
22u
100n
GND
GND
GND
+3V3
+3V3
R4
10k
SW2 RESET
A.1
D.4
GND
U1 ESP32-S3-WROOM-1-N8R2
GND.1
3V3.2
EN.3
IO4.4
IO5.5
IO6.6
IO15.8
IO16.9
IO17.10
IO18.11
IO8.12
IO19.13
IO20.14
IO3.15
IO10.18
IO11.19
IO12.20
IO13.21
IO14.22
IO21.23
IO47.24
IO48.25
IO0.27
IO35.28
IO36.29
IO37.30
IO38.31
IO39.32
IO40.33
IO41.34
IO42.35
RXD0.36
TXD0.37
IO1.39
GND.40
GND.41
+3V3
R5
10k
SW1 BOOT
A.1
D.4
GND
SPI_MOSI
TEMP_ADC
C13
1u
GND
WG_D1
GND
TAMPER
IN4
USB_DN
IN2
WG_D0
SPI_DC
SPI_SCK
SPI_CS
USB_DP
IN3
IN1
IO6
SPI_BL_BUSY
R8
1k
IO48
ETH_CS
I2C_SDA
RS485_RXD
TP2 RS485_RXD
1
BAT_SENSE
GND
GND
I2C_SCL
EXP_INT
RS485_DE
ETH_MISO
ETH_INT
IO40
TP3 IO40
1
RS485_DI
TP1 RS485_DI
1
R81
470
+3V3
D23
+3V3
D8

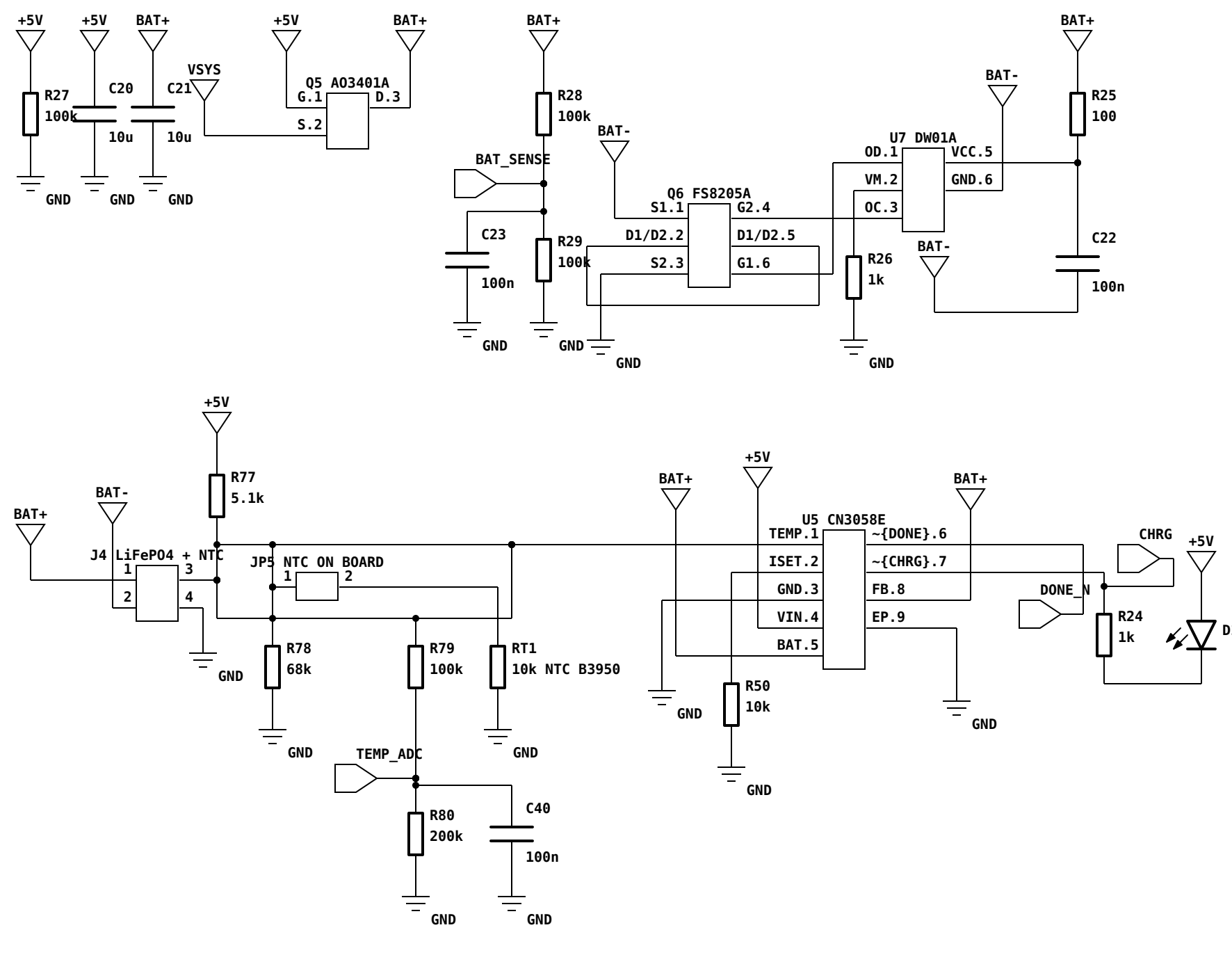

+3V3
C16
100n
GND
D1_RAW
R10
1k
+3V3
R12
10k
D6
GND
D5
GND
+3V3
R11
10k
D0_RAW
R9
1k
+3V3
U4 SN74LVC2G17
1A.1
2Y.4
GND.2
VCC.5
2A.3
1Y.6
C15
100p
GND
C14
100p
GND
GND
WG_D1
WG_D0

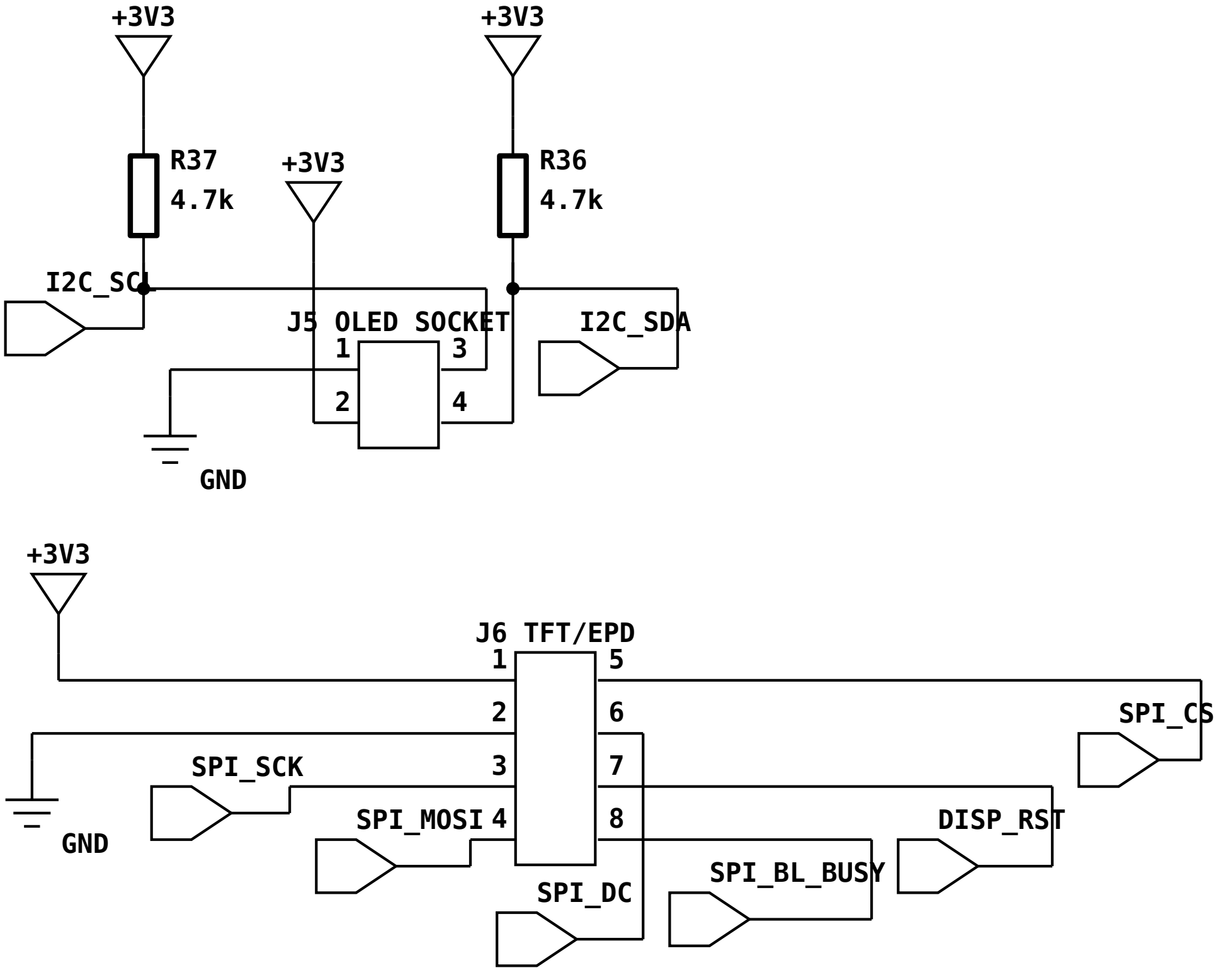

IO6
R6
100
Q1 2N7002
G.1
D.3
S.2
R13
100k
RDR_LED
GND
GND
IO40
R7
100
Q2 2N7002
G.1
D.3
S.2
RDR_BEEP
R14
100k
GND
GND
+3V3
R35
10k
SW3 TAMPER
A.1
D.4
J8 TAMPER EXT
1
2
TAMPER
GND
GND

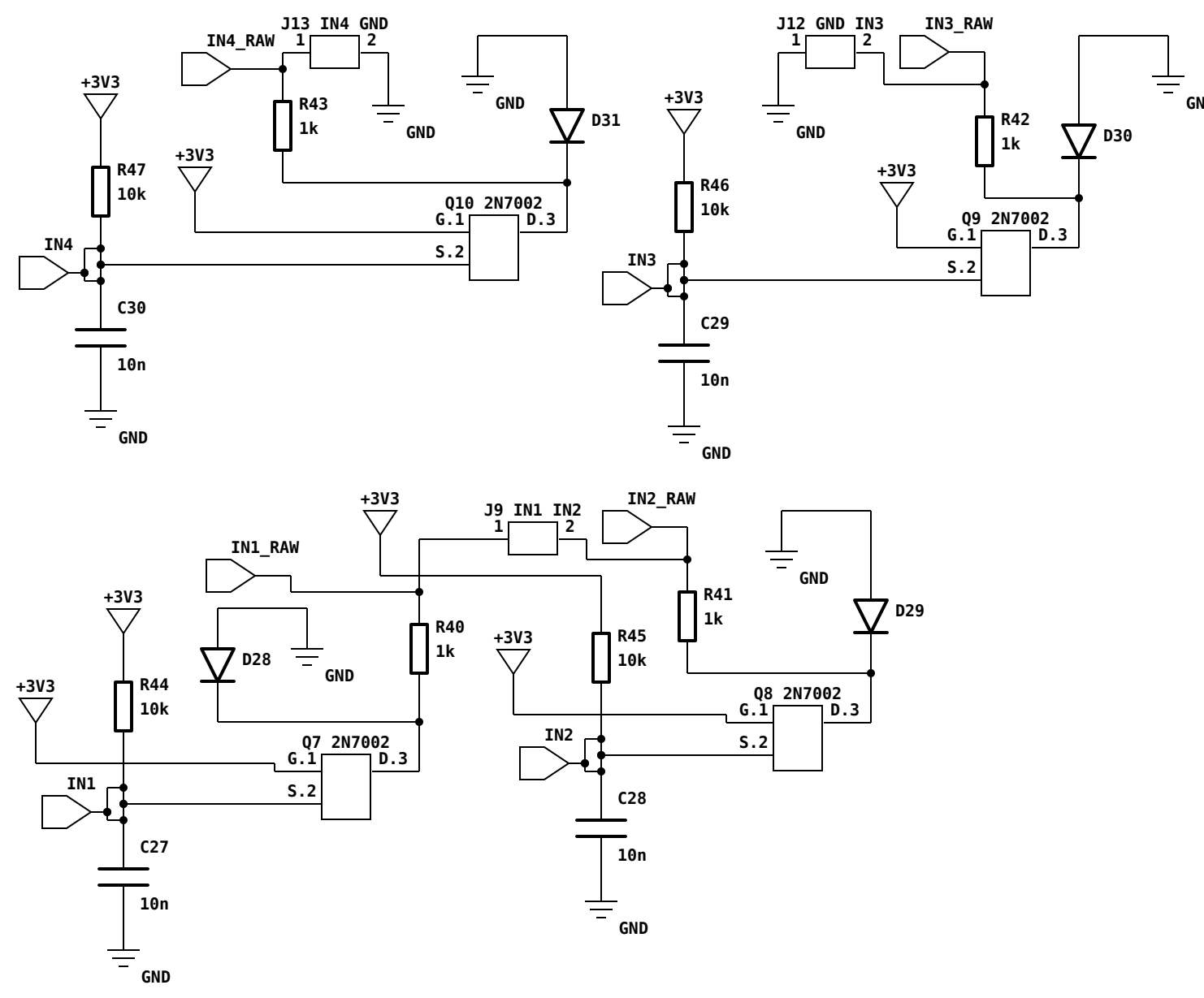

VSYS
C17
22u
GND
D0_RAW
J4 READER D0 D1
1
2
D1_RAW
RDR_LED
J11 READER LED BEEP
1
2
RDR_BEEP
BOOST_OUT
C18
22u
GND
BOOST_OUT
C19
22u
GND
+12V_RDR
J3 READER PWR
1
2
GN

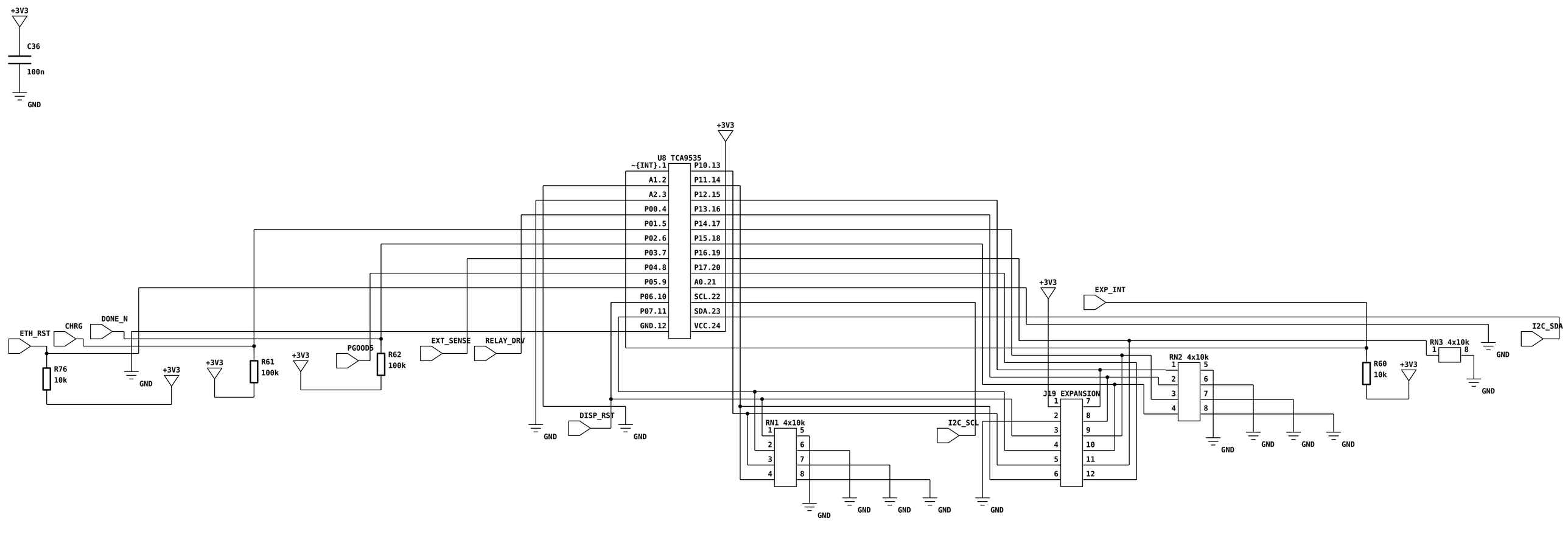

+12V_RDR
R52
100k
Q11 AO3401A
G.1
D.3
S.2
F1 0.5A
1
2
IO48
BOOST_SNS
R51
100k
R17
100
Q4 2N7002
G.1
D.3
S.2
+3V3
R18
10k
GND

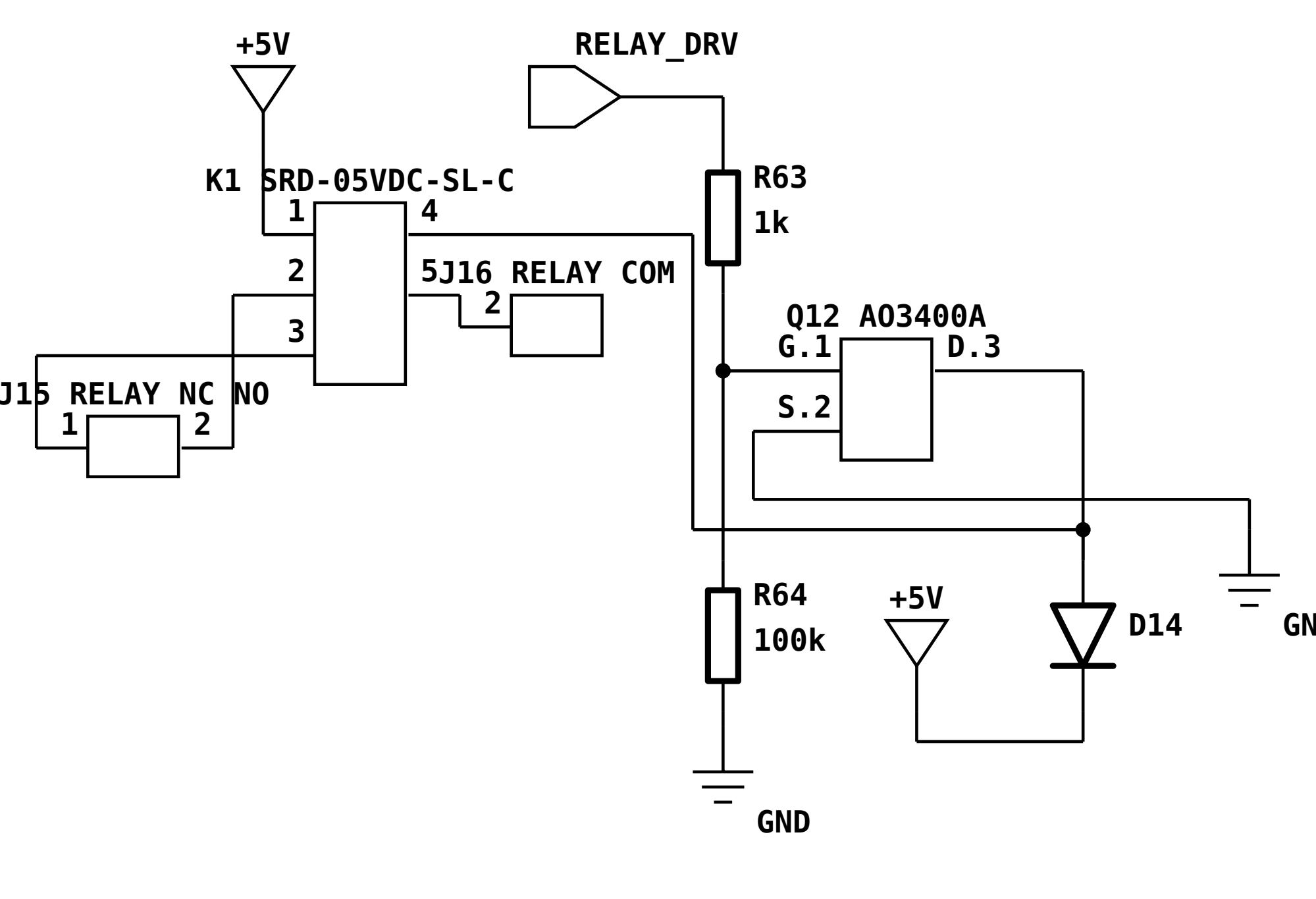

VSYS
BOOST_OUT
VSYS
L3
4.7uH
R19
200k
R21
100k
VSYS
U6 MT3608
SW.1
EN.4
GND.2
IN.5
FB.3
OOST_OUT
D10
R20
10k
R30
200k
GND
JP1 13.2V
1
2
GND
GND

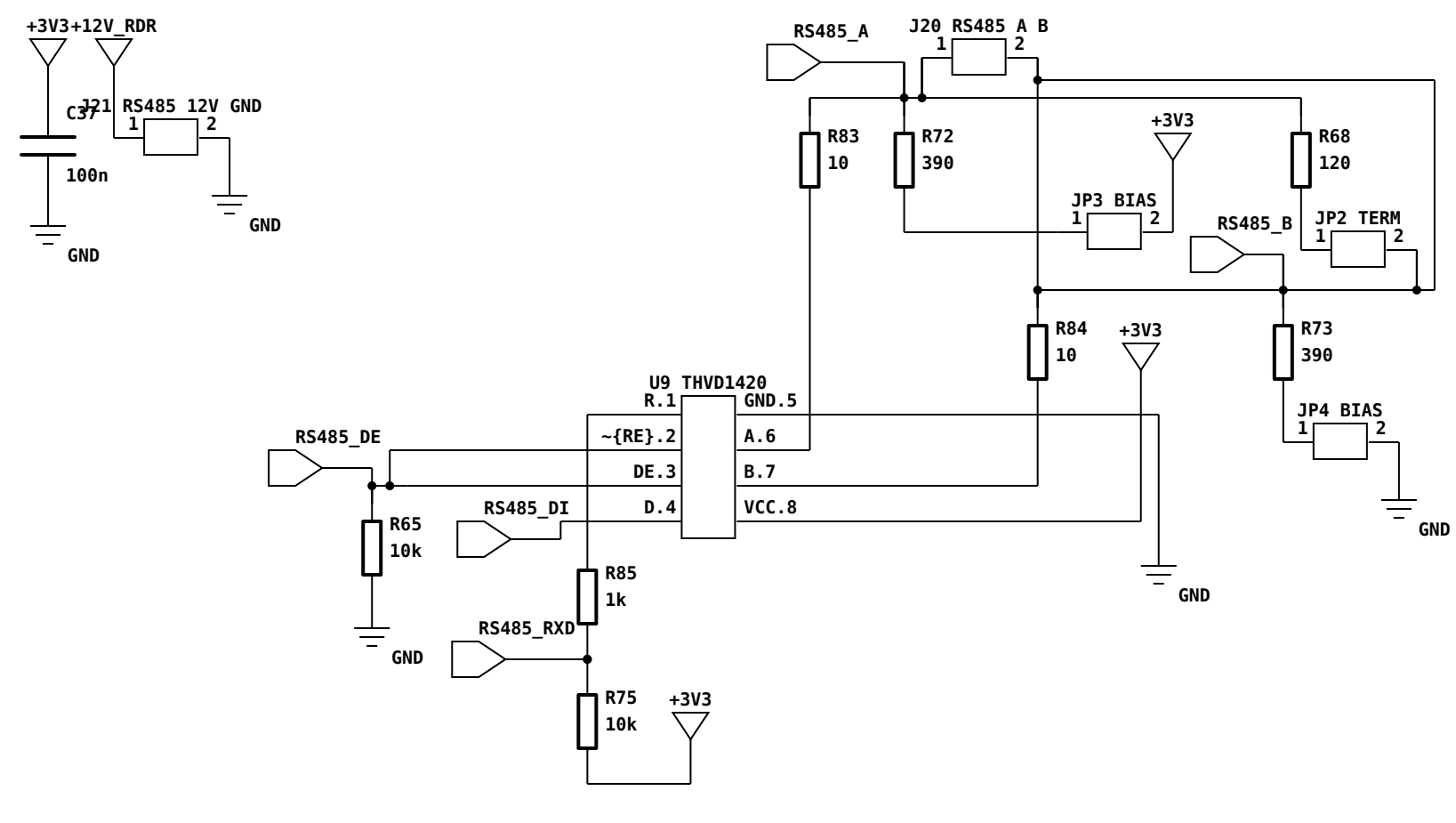

+5V
+5V
BAT+
+5V
BAT+
R27
100k
C20
10u
C21
10u
VSYS
Q5 AO3401A
G.1
D.3
S.2
GND
GND
GND
BAT+
BAT-
BAT+
R28
100k
BAT_SENSE
C23
100n
R29
100k
Q6 FS8205A
S1.1
G2.4
D1/D2.2
D1/D2.5
S2.3
G1.6
GND
GND
GND
BAT-
U7 DW01A
OD.1
VCC.5
VM.2
GND.6
OC.3
R26
1k
GND
BAT+
R25
100
BAT-
C22
100n
+5V
R77
5.1k
BAT-
BAT+
J4 LiFePO4 + NTC
1
2
3
4
JP5 NTC ON BOARD
1
2
R78
68k
R79
100k
RT1
10k NTC B3950
GND
GND
GND
TEMP_ADC
R80
200k
C40
100n
GND
GND
+5V
BAT+
BAT+
U5 CN3058E
TEMP.1
ISET.2
GND.3
VIN.4
BAT.5
~{DONE}.6
~{CHRG}.7
FB.8
EP.9
R50
10k
GND
GND
CHRG
+5V
DONE_N
R24
1k
D

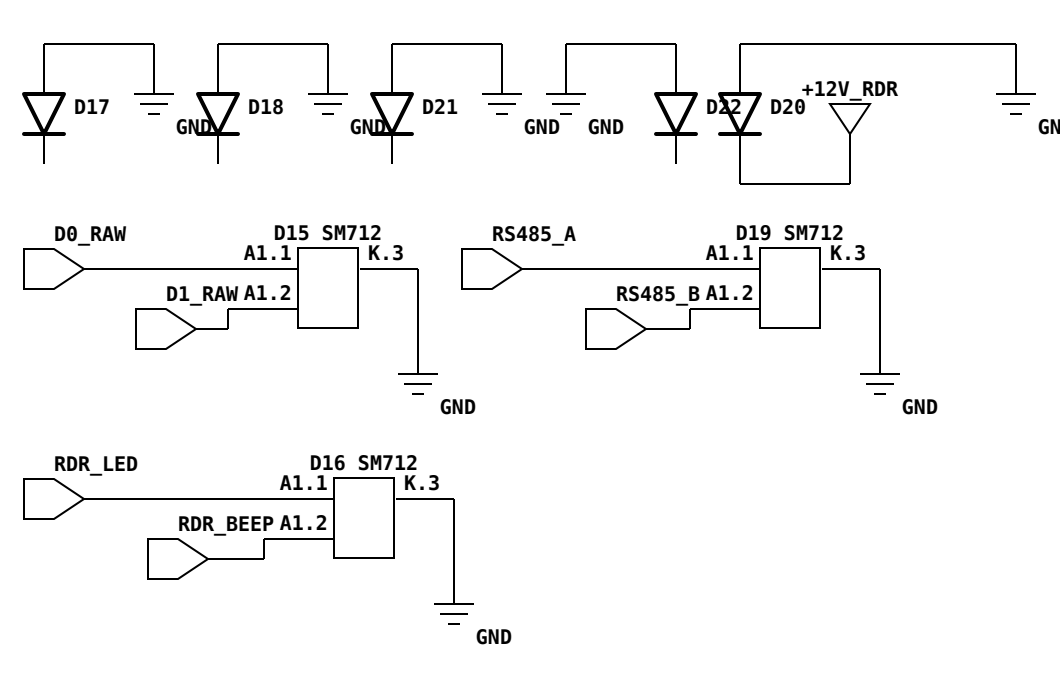

+3V3
+3V3
R37
4.7k
+3V3
R36
4.7k
I2C_SCL
J5 OLED SOCKET
1
3
2
4
I2C_SDA
GND
+3V3
J6 TFT/EPD
1
5
2
6
SPI_CS
SPI_SCK
3
7
GND
SPI_MOSI
4
8
DISP_RST
SPI_BL_BUSY
SPI_DC

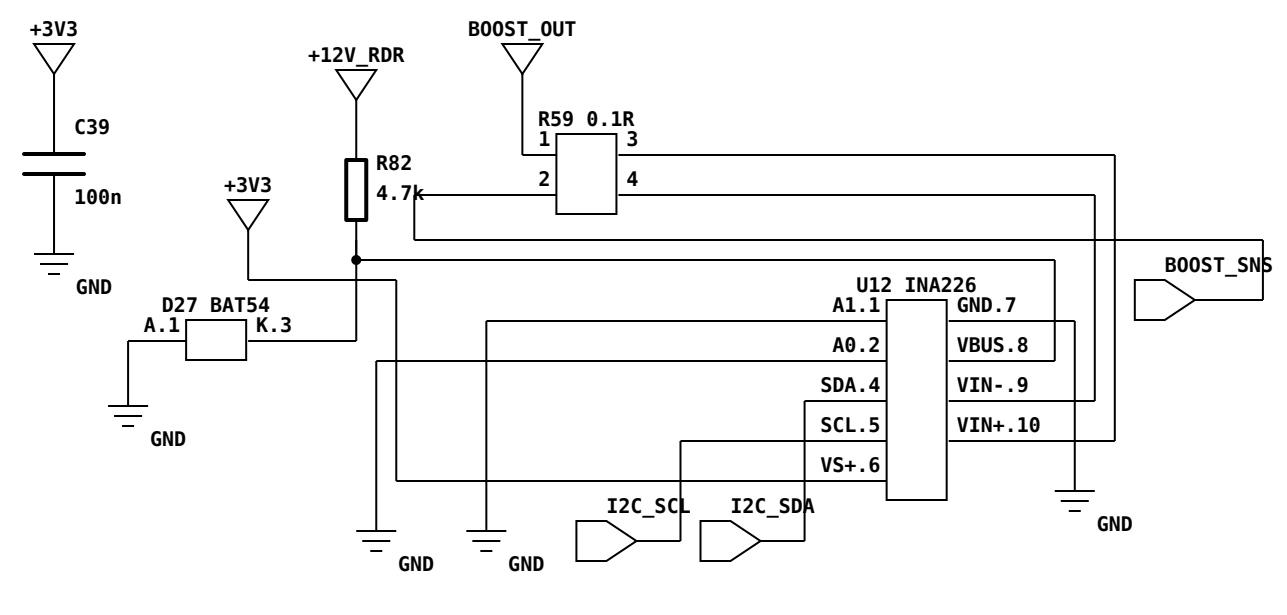

J13 IN4 GND
1
2
IN4_RAW
R43
1k
GND
GND
D31
+3V3
R47
10k
+3V3
Q10 2N7002
G.1
D.3
S.2
IN4
C30
10n
GND
J12 GND IN3
1
2
IN3_RAW
GND
R42
1k
D30
GN
+3V3
Q9 2N7002
G.1
D.3
S.2
+3V3
R46
10k
IN3
C29
10n
GND
J9 IN1 IN2
1
2
IN2_RAW
IN1_RAW
+3V3
GND
R41
1k
D29
R40
1k
D28
GND
+3V3
R45
10k
+3V3
R44
10k
Q8 2N7002
G.1
D.3
S.2
Q7 2N7002
G.1
D.3
S.2
IN2
IN1
C28
10n
C27
10n
GND
GND

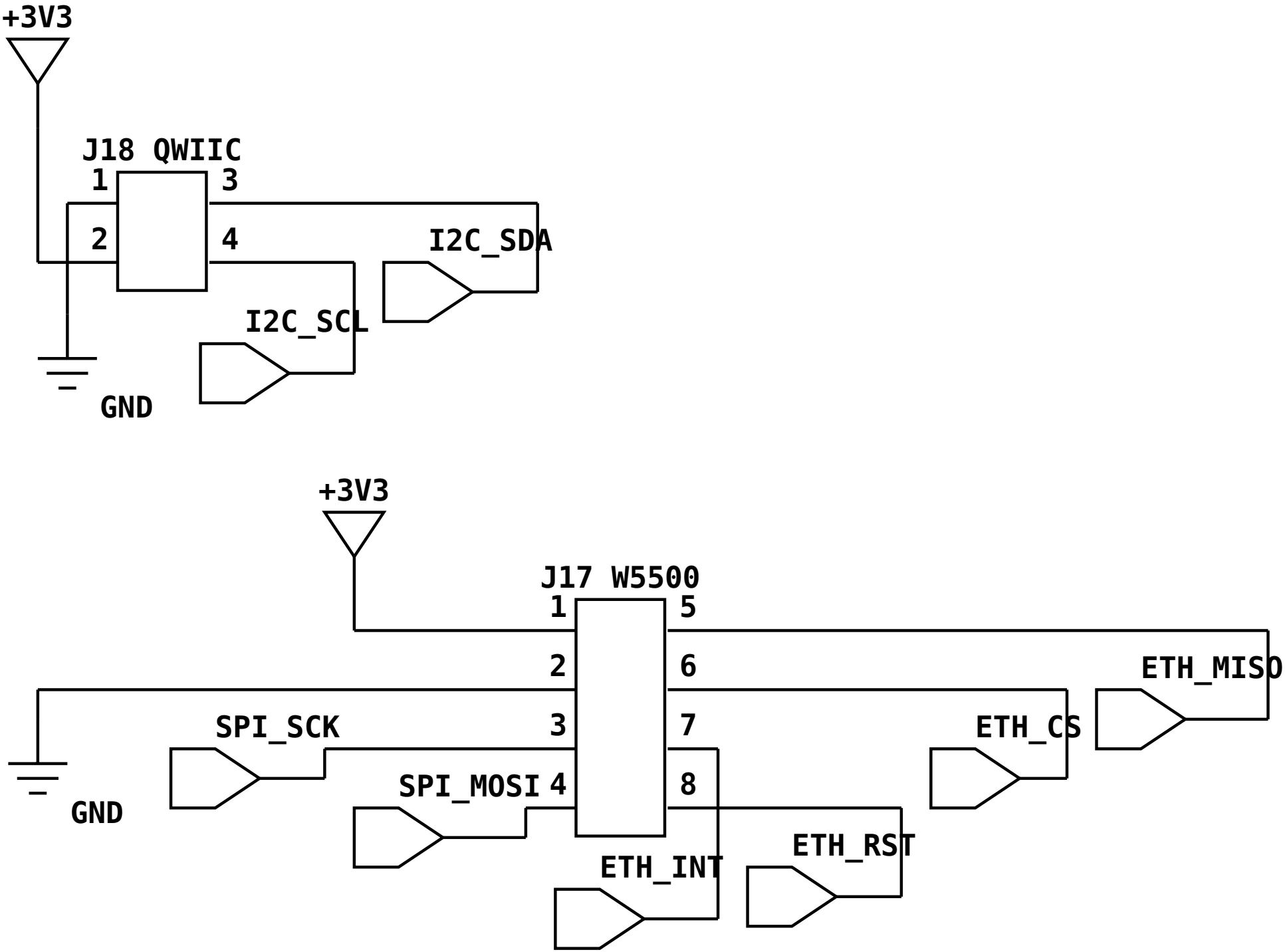

+3V3
C36
100n
GND
U8 TCA9535
~{INT}.1
A1.2
A2.3
P00.4
P01.5
P02.6
P03.7
P04.8
P05.9
P06.10
P07.11
GND.12
P10.13
P11.14
P12.15
P13.16
P14.17
P15.18
P16.19
P17.20
A0.21
SCL.22
SDA.23
VCC.24
ETH_RST
CHRG
DONE_N
R76
10k
R61
100k
PGOOD5
R62
100k
EXT_SENSE
RELAY_DRV
DISP_RST
RN1 4x10k
I2C_SCL
J19 EXPANSION
EXP_INT
RN2 4x10k
R60
10k
RN3 4x10k
I2C_SDA


+5V
RELAY_DRV
K1 SRD-05VDC-SL-C
1
4
2
3
5J16 RELAY COM
2
J15 RELAY NC NO
1
2
R63
1k
Q12 AO3400A
G.1
D.3
S.2
R64
100k
+5V
D14
GND
GN


+3V3
+12V_RDR
C37
100n
GND
J21 RS485 12V GND
1
2
GND
RS485_A
J20 RS485 A B
1
2
R83
10
R72
390
+3V3
JP3 BIAS
1
2
R68
120
RS485_B
JP2 TERM
1
2
R84
10
+3V3
R73
390
JP4 BIAS
1
2
GND
U9 THVD1420
R.1
GND.5
~{RE}.2
A.6
DE.3
B.7
D.4
VCC.8
RS485_DE
RS485_DI
R65
10k
GND
R85
1k
RS485_RXD
R75
10k
+3V3
GND


D17
GND
D18
GND
D21
GND
GND
D22
D20
+12V_RDR
GN
D0_RAW
D15 SM712
A1.1
K.3
D1_RAW
A1.2
GND
RS485_A
D19 SM712
A1.1
K.3
RS485_B
A1.2
GND
RDR_LED
D16 SM712
A1.1
K.3
RDR_BEEP
A1.2
GND


+3V3
C39
100n
GND
+12V_RDR
R82
4.7k
+3V3
D27 BAT54
A.1
K.3
GND
BOOST_OUT
R59 0.1R
1
3
2
4
U12 INA226
A1.1
GND.7
A0.2
VBUS.8
SDA.4
VIN-.9
SCL.5
VIN+.10
VS+.6
BOOST_SNS
I2C_SCL
I2C_SDA
GND
GND
GND


+3V3
J18 QWIIC
1
3
2
4
I2C_SDA
I2C_SCL
GND
+3V3
J17 W5500
1
5
2
6
ETH_MISO
SPI_SCK
3
7
ETH_CS
GND
SPI_MOSI
4
8
ETH_RST
ETH_INT